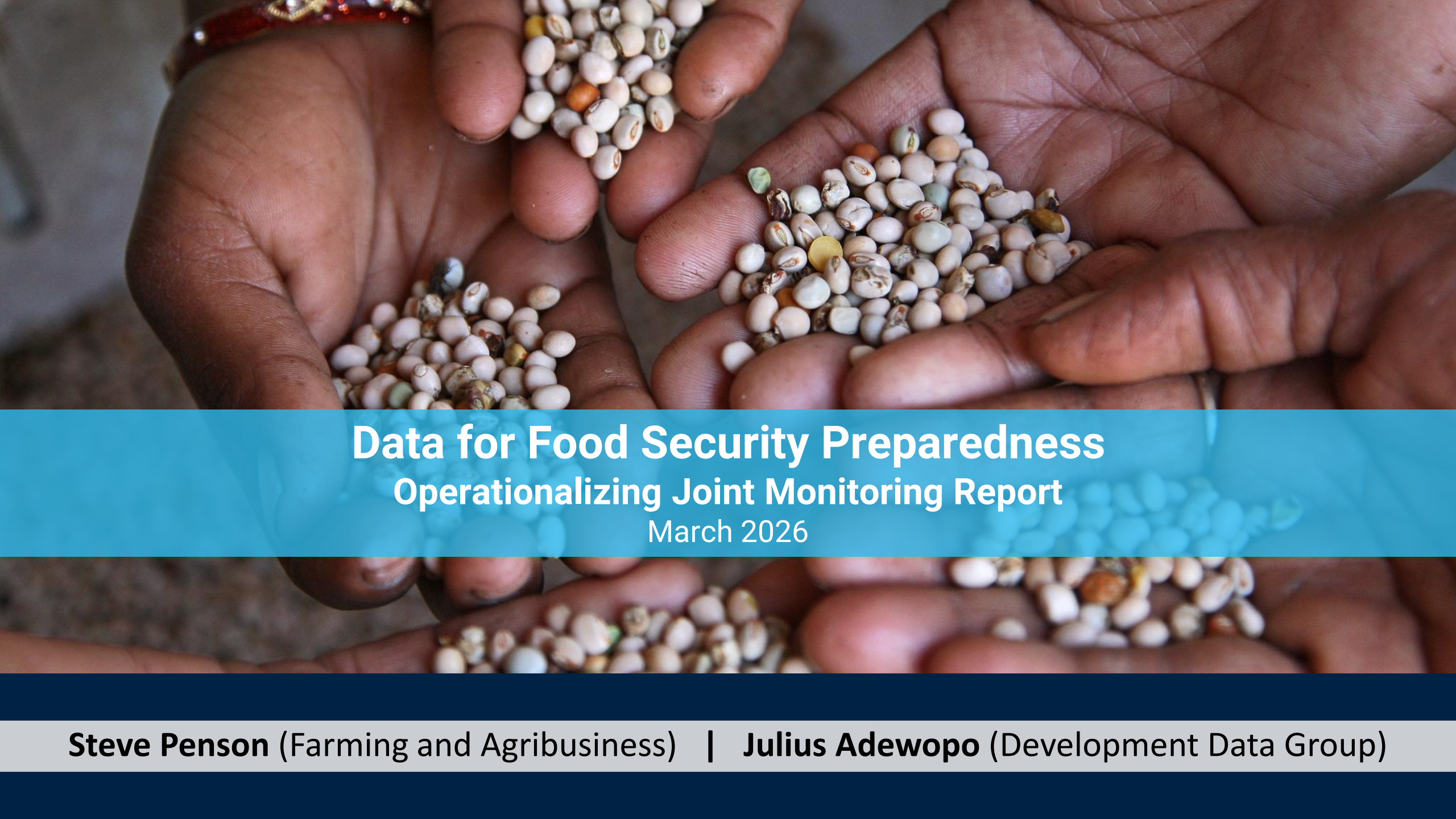


A close-up photograph of several hands holding a large quantity of small, light-colored seeds, possibly cowpeas, which are a staple food crop in many developing regions. The hands are of various skin tones, suggesting a diverse group of people. The seeds are piled in the palms and fingers, with some spilling out. The background is blurred, focusing attention on the hands and the seeds.

Data for Food Security Preparedness

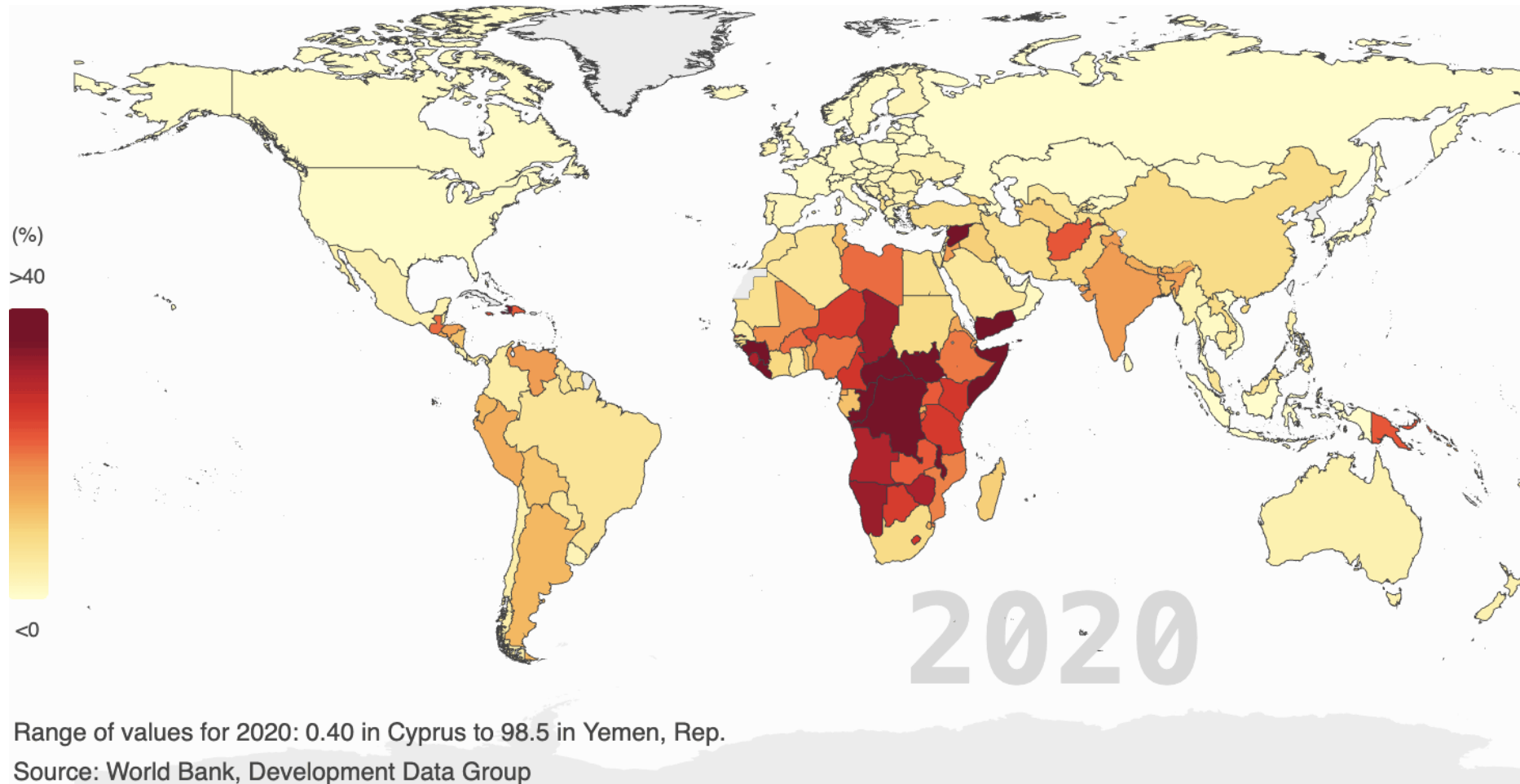
Operationalizing Joint Monitoring Report

March 2026

Steve Penson (Farming and Agribusiness) | **Julius Adewopo** (Development Data Group)

|| Hunger Is on the Rise – How Do We Respond?

Global Projected Prevalence of Severe Food Insecurity



Food Security Preparedness Planning

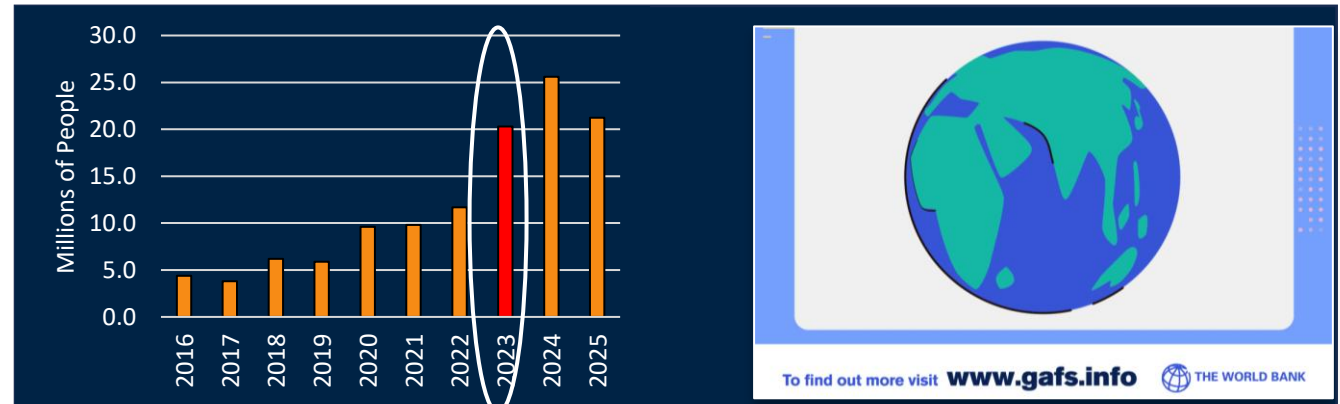
AVERTING FOOD AND NUTRITION SECURITY CRISES

A NEW APPROACH FOR ADVANCING EARLY AND IMPACTFUL RESPONSES AT SCALE

PREPAREDNESS
PLAN FOR FOOD
AND NUTRITION
SECURITY CRISES
QUALITY ASSURANCE CRITERIA

GLOBAL
LEADERSHIP
ROUNDTABLE
SUPPORT TEAM
OPERATIONS MANUAL

The Preparedness Plan addresses extraordinary food and nutrition crises extending beyond chronic conditions

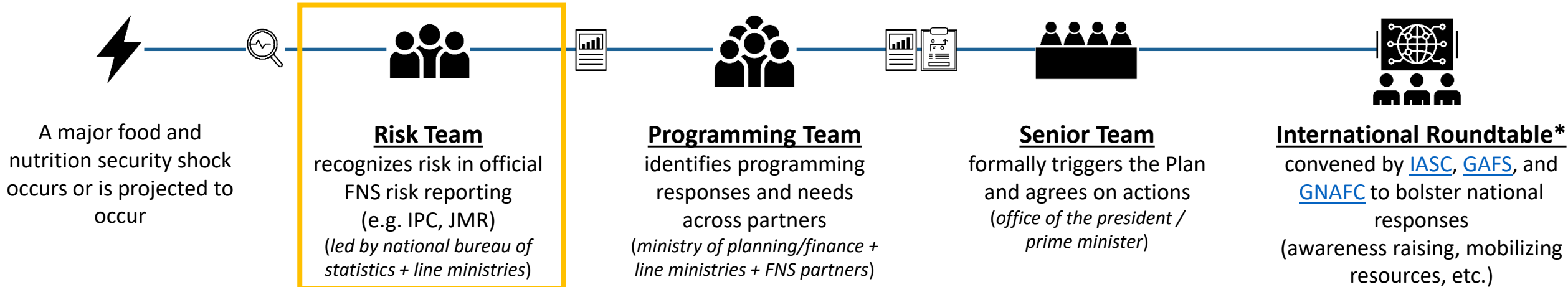


For illustrative purposes

How the Plan is unique:

- It is a National Plan (government owned and led)
- Seeks to mitigate extreme fluctuations
- It focuses on mobilizing collective and early attention and action from all FNS stakeholders (hum + dev nexus)
- It focuses on evidence-based decision-making protocols
- It creates a foundation for future government ownership / leadership (capacity and institution building)

How Preparedness Plans work

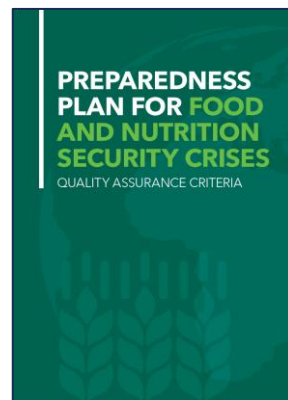


Additional Background

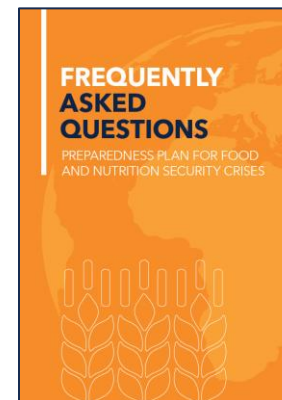
Briefing Note



Quality Assurance



FAQs



Operational Terms



|| Identifying When Major Shocks Occur

- ● ● ● **Chronic / protracted needs** →
- Humanitarian responses (e.g. HNO/HRP)
 - Development (e.g. core country envelopes)
 - Food Systems Transformation / Dialogue Agenda
-

- ⚡ ●● **Major shocks → FNS crises** →
- When a shock occurs, there is a need to mobilize scaled up, early, and collective responses (Gov + Hum + Dev)

|| Joint Monitoring Report (JMR)

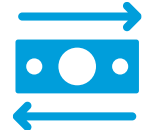
- JMR is a light weight, high frequency approach food security monitoring tool.
- Through 5-9 indicators, possible to identify around 80% of historic deteriorations in food security.
- Complements both IPC and in field knowledge - model is a tool to support analysis.
- Opportunities to support IPC/CH enhanced risk analysis efforts.
- JMR is being operationalized as an element of preparedness plan triggering mechanism.



Food Prices



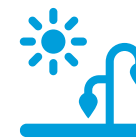
Market Volatility



Exchange rate



Conflict



Drought



Food imports

|| Joint Monitoring Report (JMR)



Joint / collaborative effort



Consolidation of existing information across partners



User friendly (e.g. 5 pgs)



Complementary to major IPC reports



Integration of select, robust, and well-vetted data

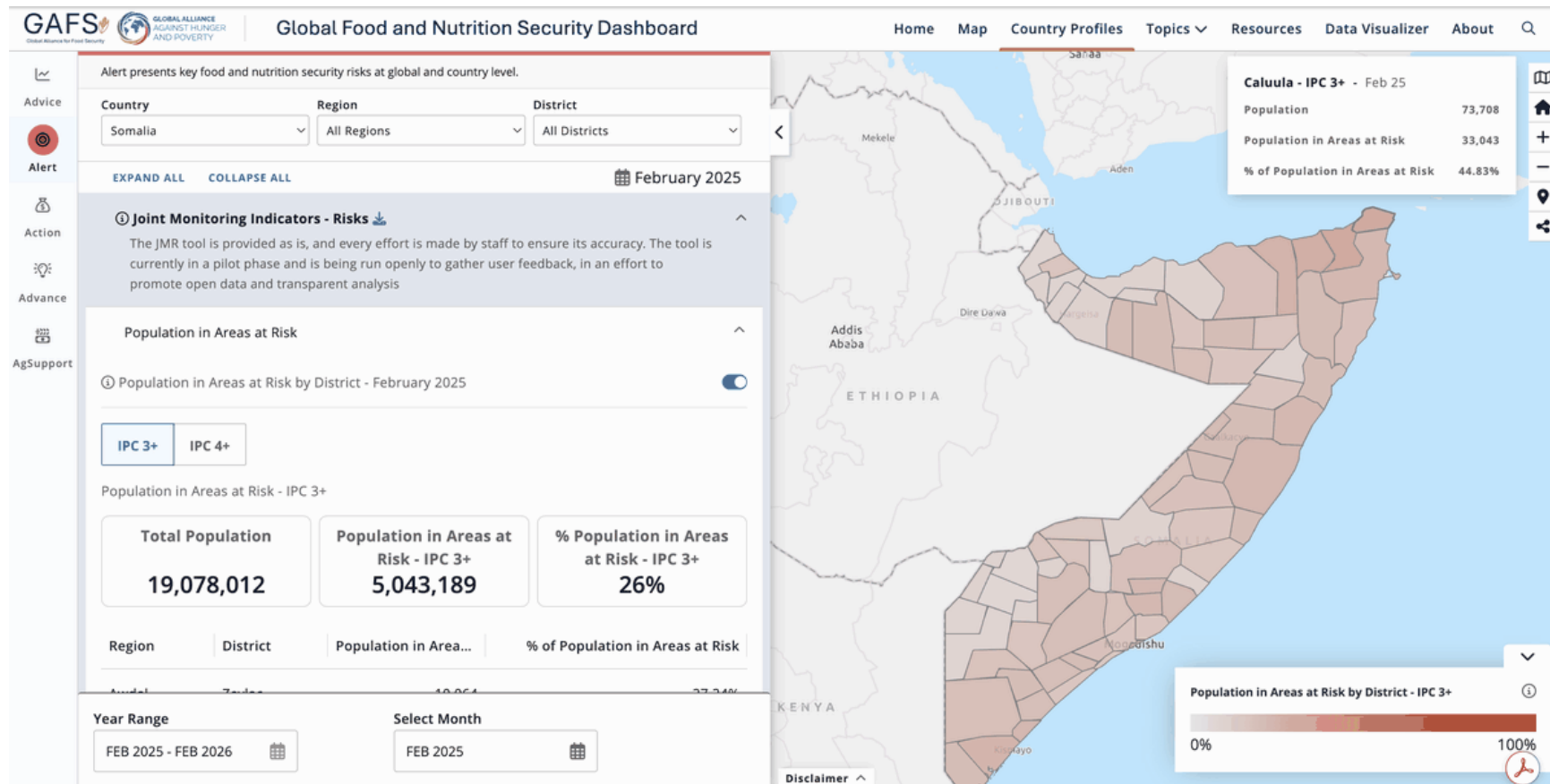


Higher frequency reporting (bimonthly / quarterly)



Intended for decision-makers and to inform FSCPP activation

Global Food and Nutrition Security Dashboard



- JMR integrated into the Global Food and Nutrition Security Dashboard.
- Available through: <https://www.gafrs.info/jmr>

Joint Monitoring Report Coverage

Joint Monitoring Lead	Country
Government led	Chad
	Democratic Republic of the Congo
	Ethiopia
	Haiti
	Liberia
	Mozambique
	Niger
	Sierra Leone
	Burkina Faso
	Cameroon
	Central African Republic
	Madagascar
	Malawi
	Nigeria
International led	Sudan
	Yemen



Country Risk Teams – A Truly Joint Collaboration

In-country risk team engagement with government and international teams



Regular and consistent Joint Monitoring Reports



Country Risk Teams – A Truly Joint Collaboration

Yemen JMR drafting team



Sudan JMR drafting team



Nigeria JMR drafting team



Somalia JMR drafting team



Cameroon JMR drafting team



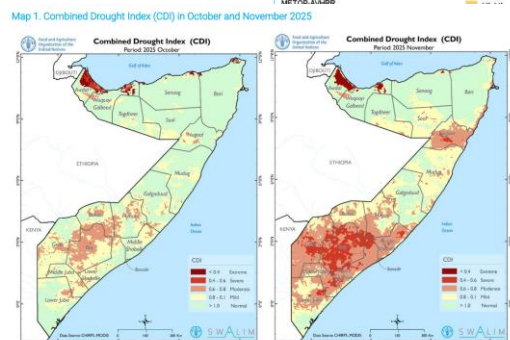
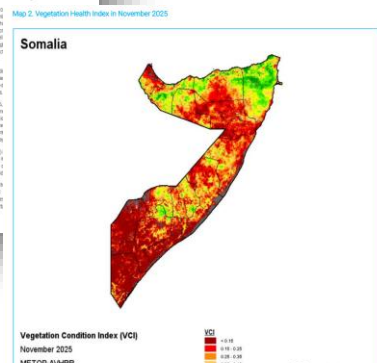
NOTE: example members,
not all members included

|| CASE STUDY: Triggering of the Somalia Preparedness Plan

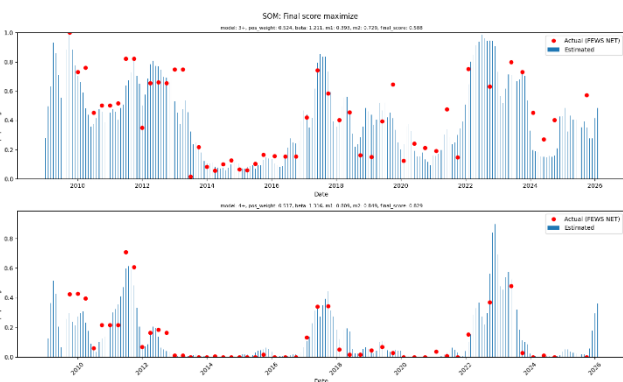
Abdullahi Kelly, Somalia Nation Bureau of Statistics



CASE STUDY: Triggering of the Somalia Preparedness Plan



Source: SWALIM



23 Dec 2025
Risk team releases
Somalia Joint
Monitoring Report #4

24 Dec 2025
Programming team
recommend triggering
based on risk team
pre-assessment

January 2026
Preparedness plan
formally triggered by
the Prime Minister

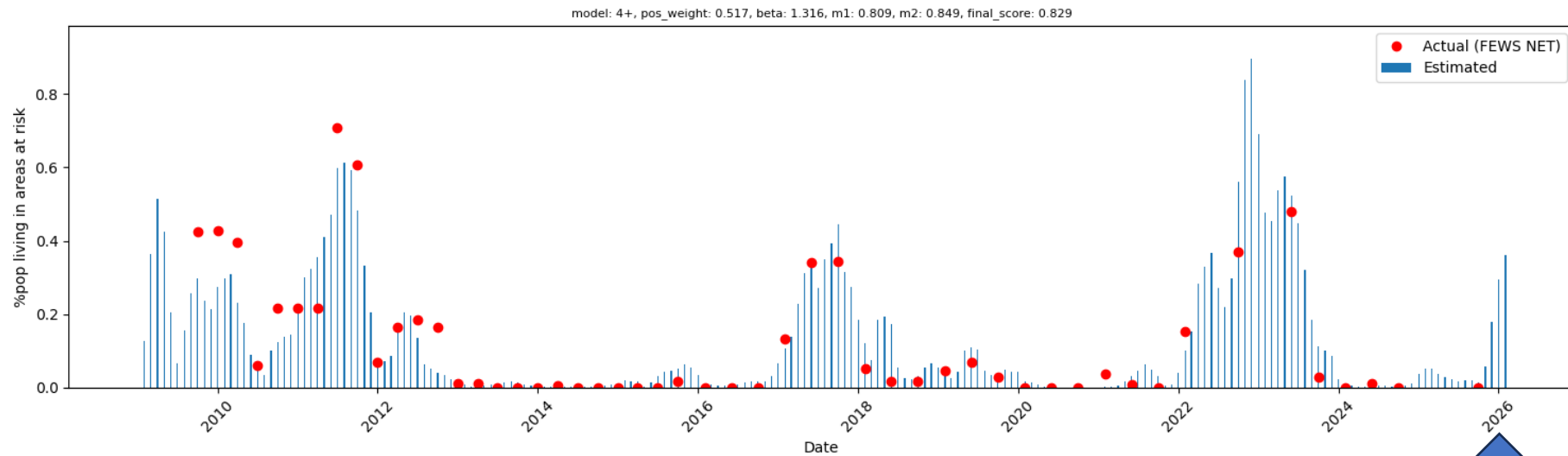
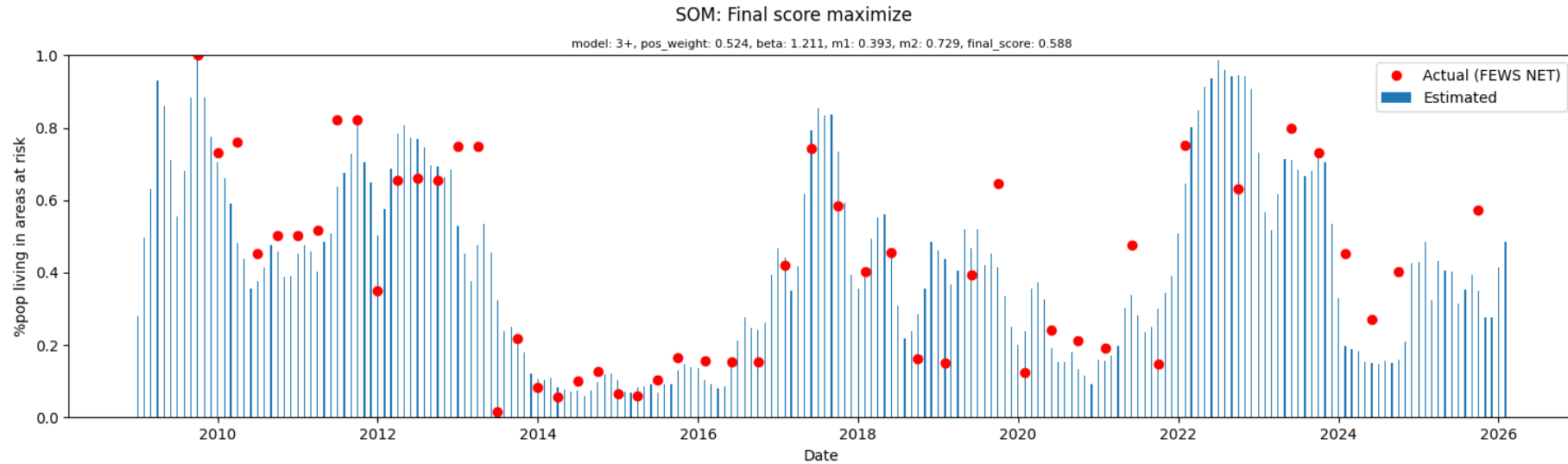
26 Feb 2026
International round
table. Development
and humanitarian
funding pledged.



EU announces €63 million for Somalia
to respond to the worsening
humanitarian crisis



|| CASE STUDY: Triggering of the Somalia Preparedness Plan

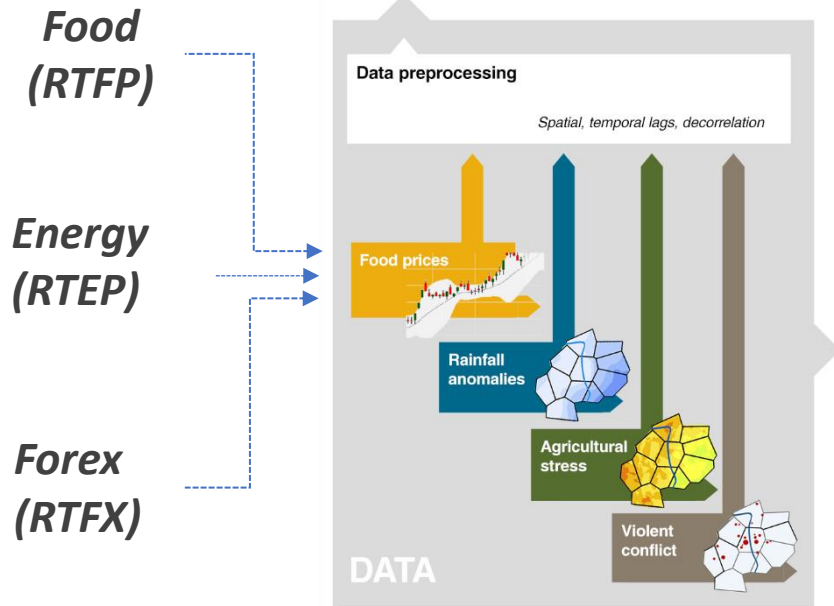


Triggering of the preparedness plan



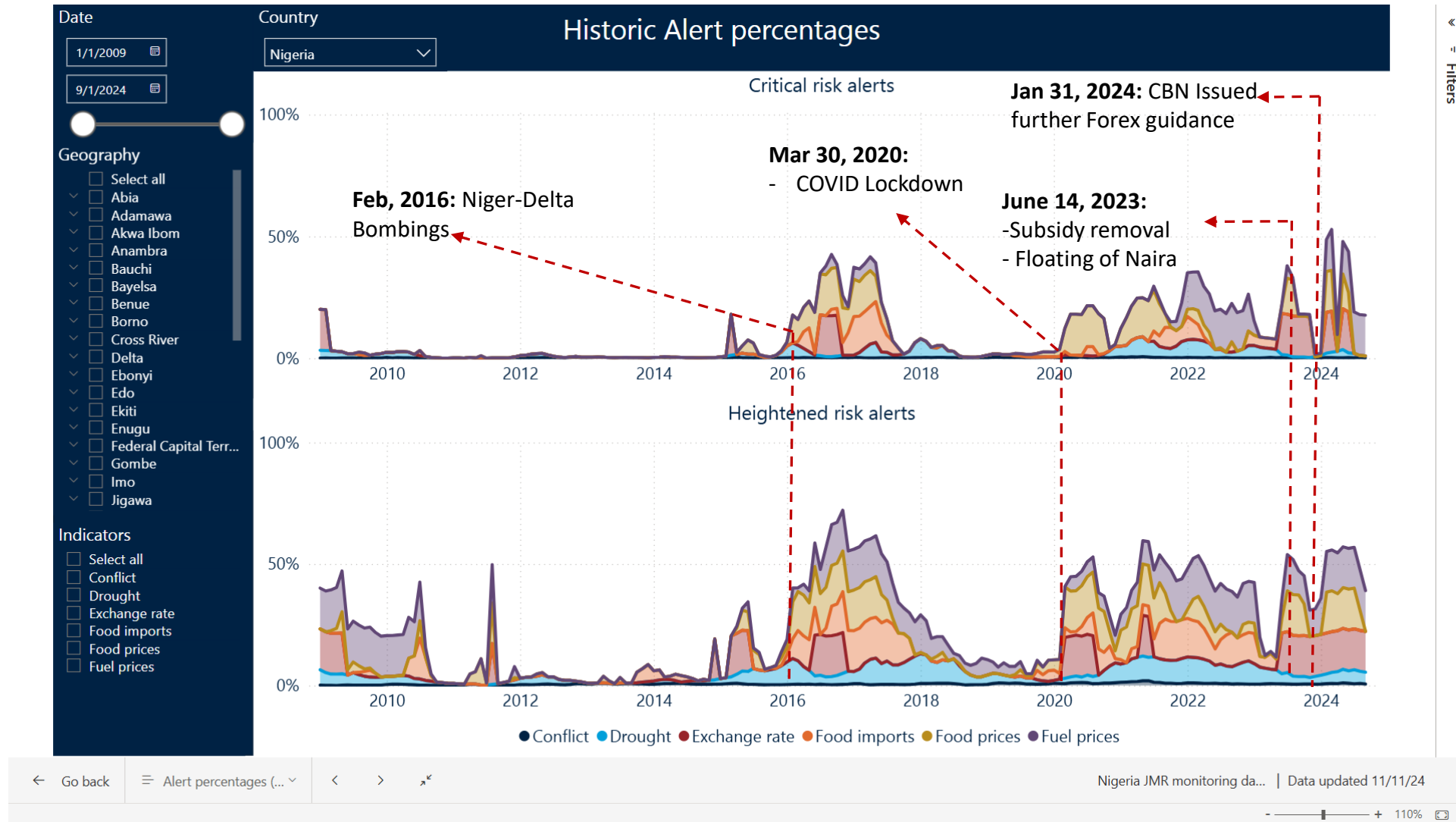
**Advancing Data System and
Tools for Timely Intelligence.**

|| JMR Risk Indicators: **Driven by Thresholds**

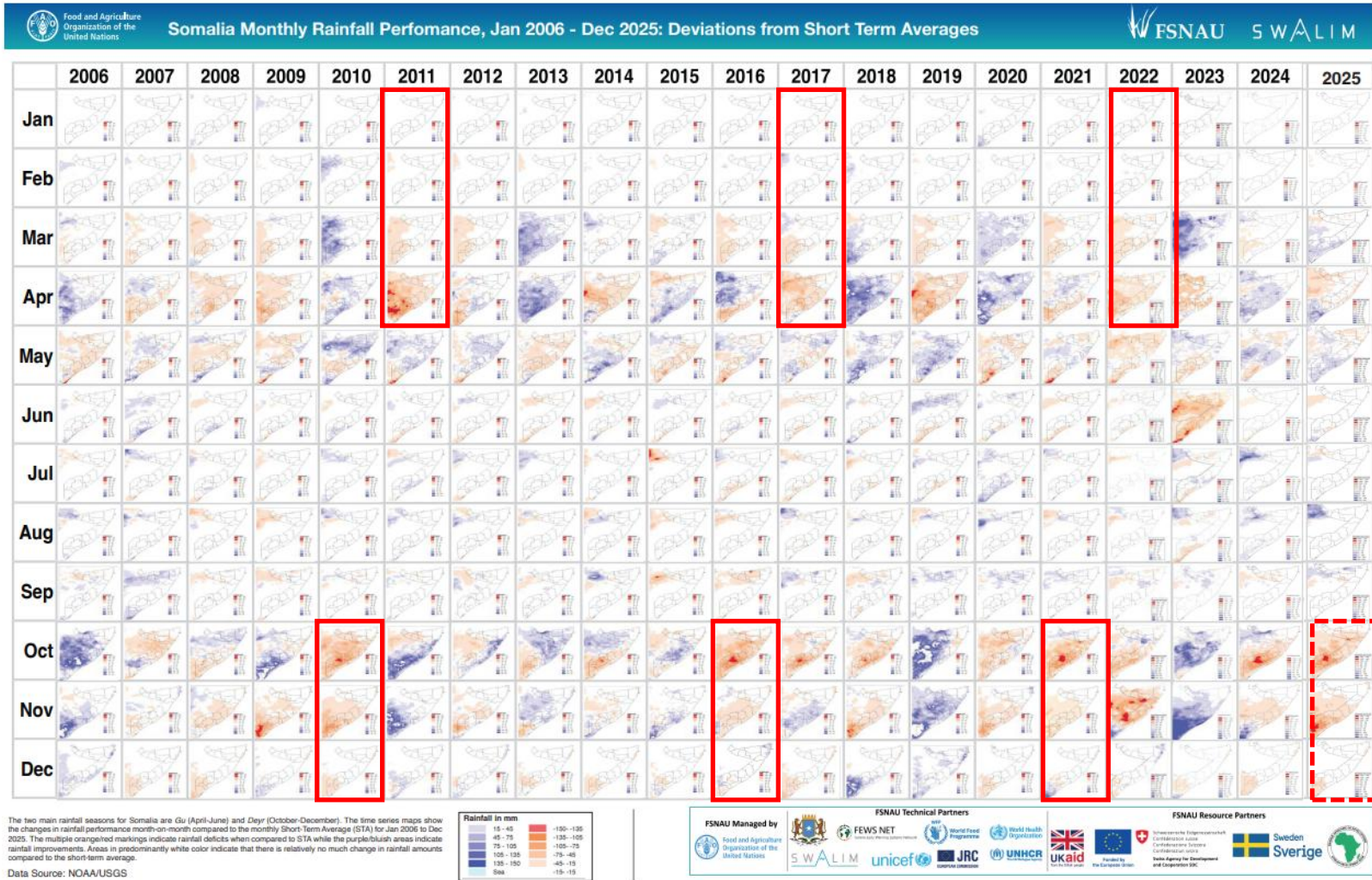


Risk Indicator	Threshold	Typical	Heightened	Critical
Minimum Food Expenditure Basket Price	% change with rolling average	<14.4% increase	14.4%-22.9% increase	>22.9% increase
Fuel Prices	Relative strength index (RSI)	<76.18	76.18-89.68	>89.68
Exchange rate	Relative strength index (RSI)	<68.26	68.26-83.08	>83.08
Conflict in neighbouring districts	Relative strength index of the Exponential Moving Average of fatality rate	<57.56	57.56-86.09	>86.09
Displacement	Relative strength index of the Exponential Moving Average of displaced	<55.86	55.86-64.06	>64.06
Drought	Average of the anomaly rates	>84%	70%-84%	<70%

JMR Risk Indicators: Driven by Thresholds e.g. Nigeria



|| JMR Risk Indicators: Driven by Thresholds e.g. Somalia



Defining “an extraordinary food and nutrition security crisis” (Somalia Prep Plan, pg. 4)

General criteria have been met

- ✓ 1. An acute shock(s) has occurred
- ✓ 2. A minimum level of FNS severity has been or is expected to be met
- ✓ 3. FNS conditions are at risk of significantly worsening

Specific criteria have been met

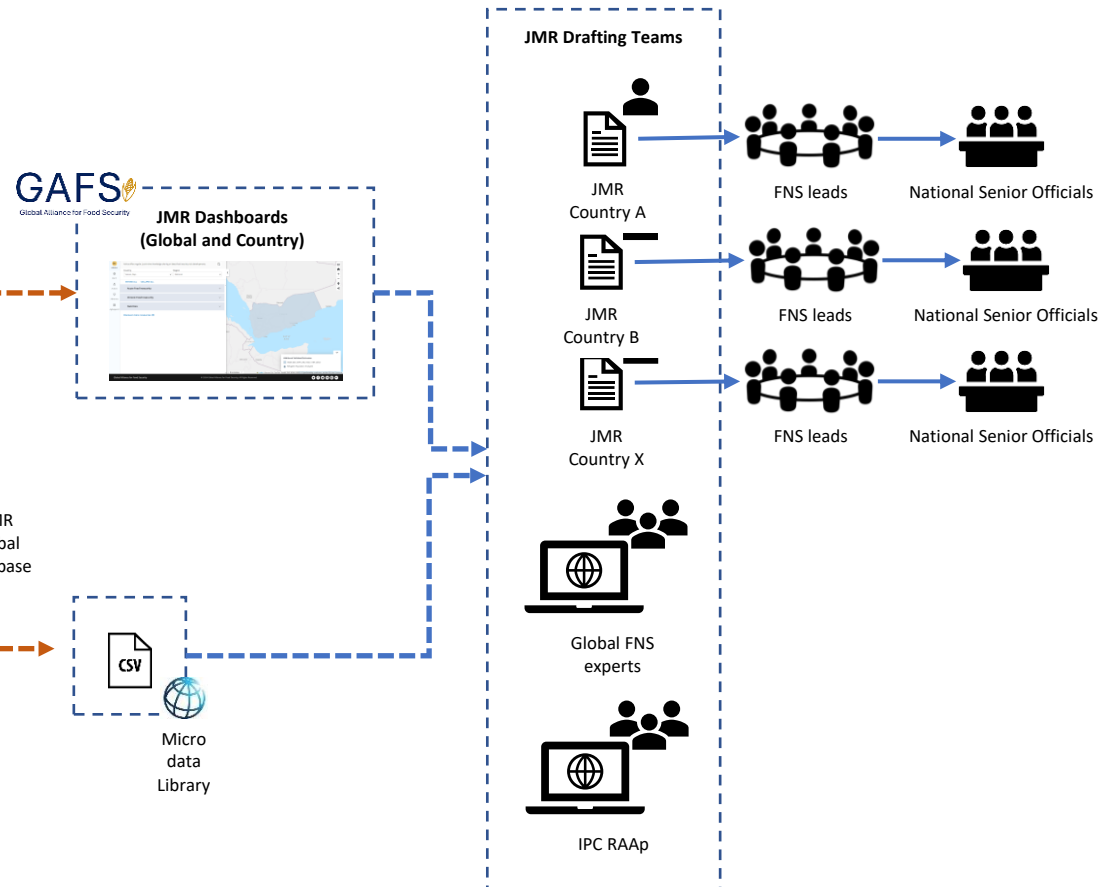
- ✓ Two consecutive failed rains accompanied by a forecasted or expected third failed rain, defined as a seasonal (Gu or Deyr) rainfall of less than 50 percent of average precipitations across two or more territories in FMS.*

Sources: FAO, FSNAU, SWALIM (2025)

* An indicator for rainfall failure is the percentage decrease below the average of long-term mean (monthly) rainfall based on Somalia Water and Land Information Management (SWALIM) monitoring.

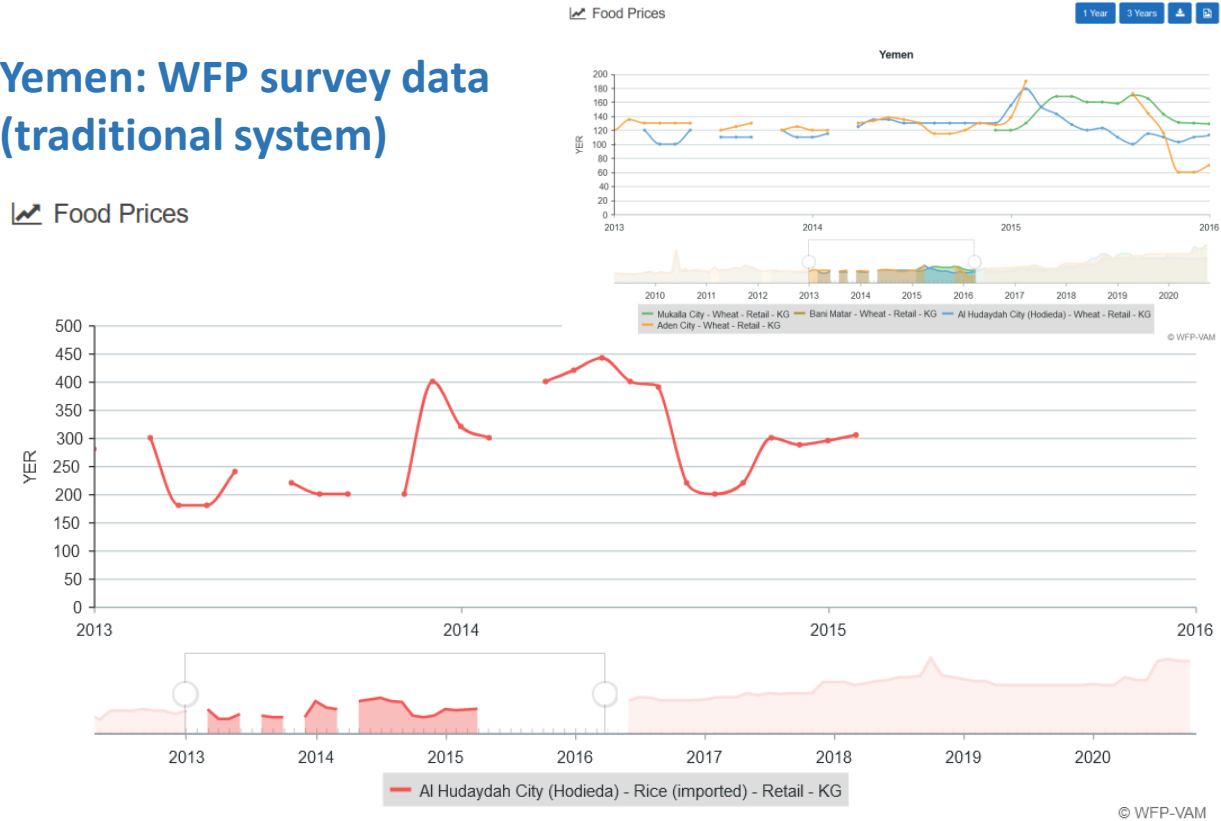
|| Innovating to Mitigate Data Gaps

- Input data, including prices, are not always reliably available across markets and time



Yemen: WFP survey data (traditional system)

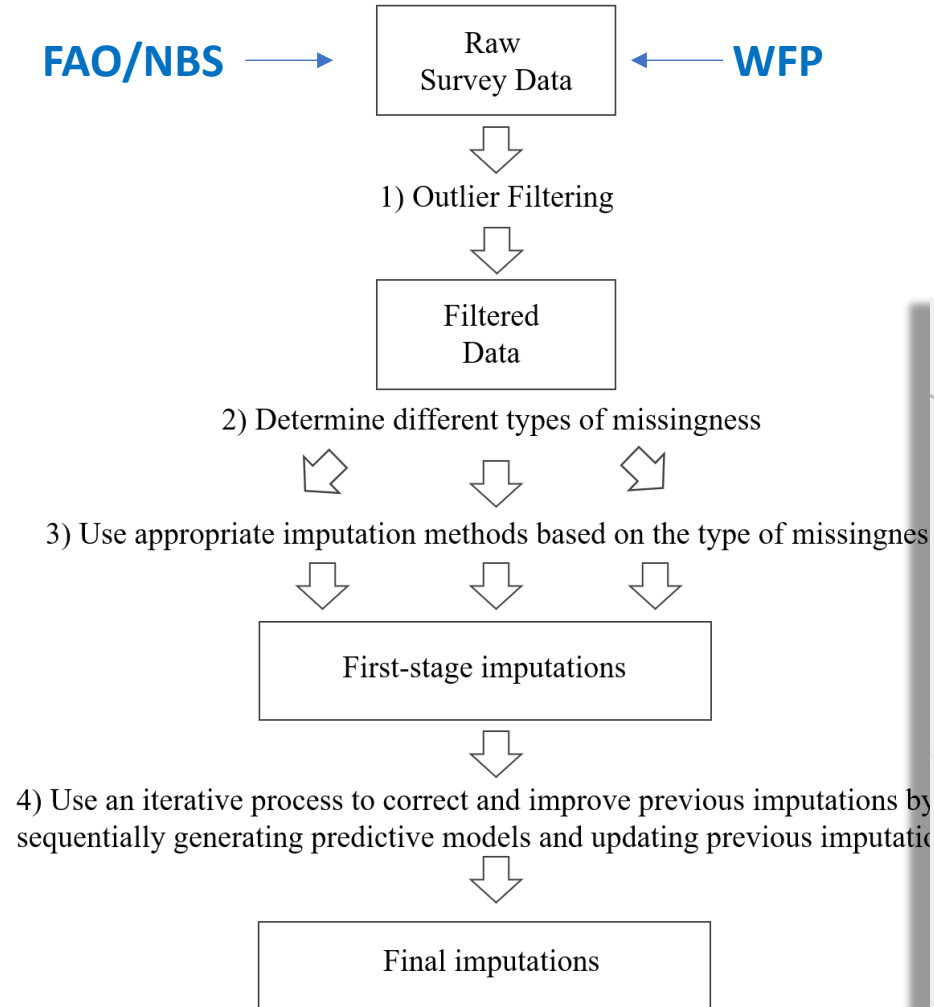
Food Prices



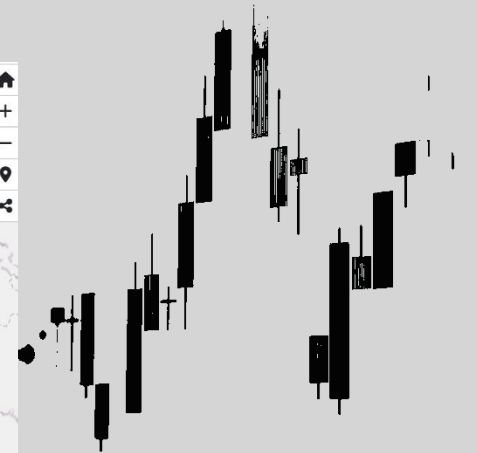
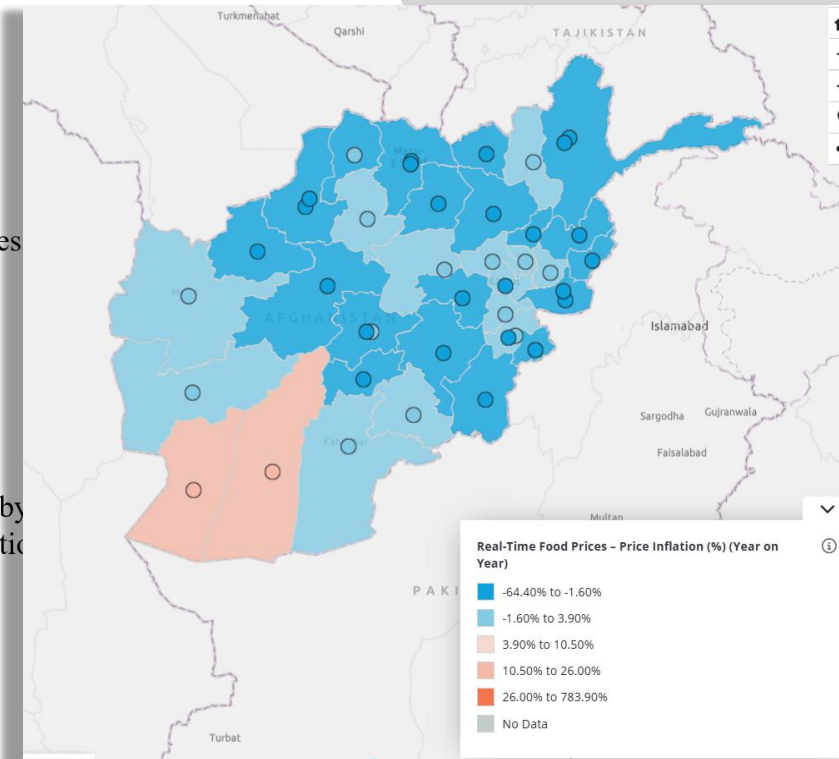
Following the Yemeni Civil War (2014–present), the Blockade of Yemen refers to a sea, land and air blockade on Yemen which started with the positioning of Saudi Arabian warships in Yemeni waters in 2015 with the Saudi Arabian-led intervention in Yemen.

Inset: Aden (port city, Saudi coalition) | Hodeida (port city, Houthi) | Sa'na (capital) **noData** | Mukalla (Al-Qaeda)

|| Innovating to Mitigate Data Gaps : **Imputation of Missing Data**



- A Fractionally Integrated **GARCH** model (Generalized Auto-Regressive Conditional Heteroskedasticity) using a Generalized Error Distribution



|| Innovating to Mitigate Data Gaps: **Real Time Price Data**

36 Countries

(LIC – MIC, incl. FCS)

**2500+ Market
Locations**

**40+
Commodities**
(unofficial exchange
rates, agricultural
wages)

**Updated
Weekly**
(now 1.4+ Million
Datapoints)

Pipeline:

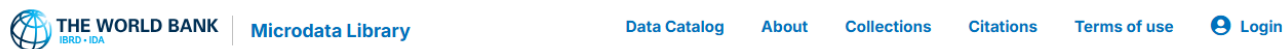
- ✓ **Automated processing:** AI-driven integration over WFP, FAO, NSO data sources.
- ✓ **Automated safeguards:** designed to operate in the event of system failures.
- ✓ **Automated data quality checks:** machine learning to fill data gaps and detect outliers.

|| Innovating to Mitigate Data Gaps: Real-time Price Data

Real Time Price (RTP) Data Pipeline

MDL - <https://microdata.worldbank.org/index.php/catalog/4483>

GAFS - https://www.gafs.info/country-profiles/?state=Alert&country=NGA&indicator=inflation_food_price_index



Home / Central Data Catalog / FCV / WLD_2021_RTFP_V02_M



Monthly food price estimates by product and market

36 countries, 2498 markets, 2007/01/01-2025/09/01, version 2025/09/29

Afghanistan, Armenia, Burundi...and 33 more, 2007 - 2025

[GET MICRODATA](#)

Reference ID WLD_2021_RTFP_v02_M

DOI <https://doi.org/10.48529/2ZH0-JF55>

Producer(s) Bo Pieter Johannes Andrée

Collection(s) [Fragility, Conflict and Violence](#)

Metadata [Documentation in PDF](#) [DDI/XML](#) [JSON](#)

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DATA DESCRIPTION

DOCUMENTATION

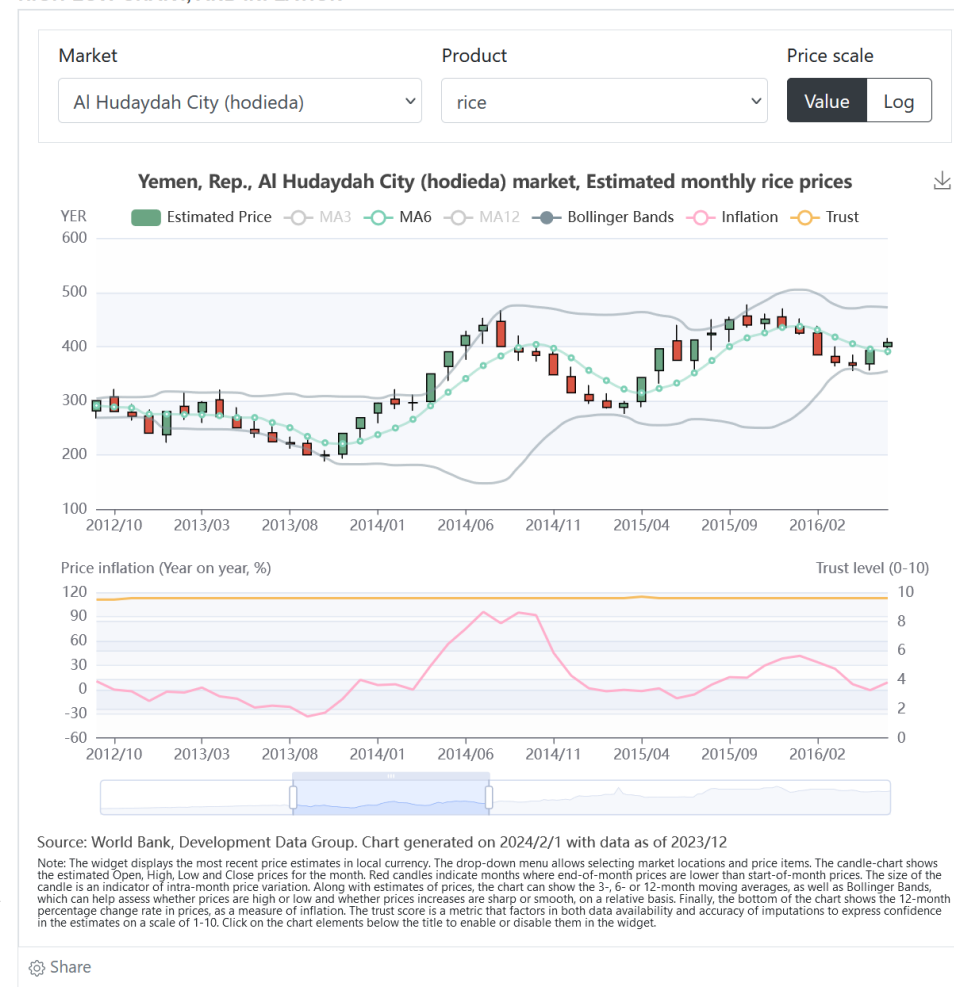
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RELATED DATASETS

DATA API

Identification

HIGH-LOW CHART, AND INFLATION



|| Innovating to Mitigate Data Gaps: Crowdsourcing Approach

High Frequency Crowdsource Price (HFCP) Data Pipeline

<https://microdata.worldbank.org/catalog/8197>



- Responsive Volunteers (778)



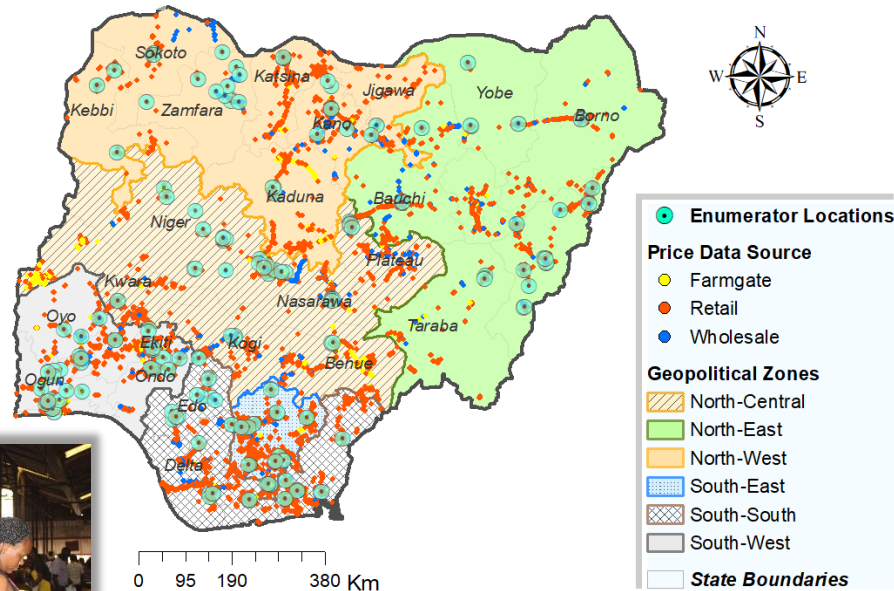
- On-time Data Flow + QC algorithm (Daily)



- Robust Data (~2.8 Million Datapoints)



- Scalable (10 Commodities 36 States)



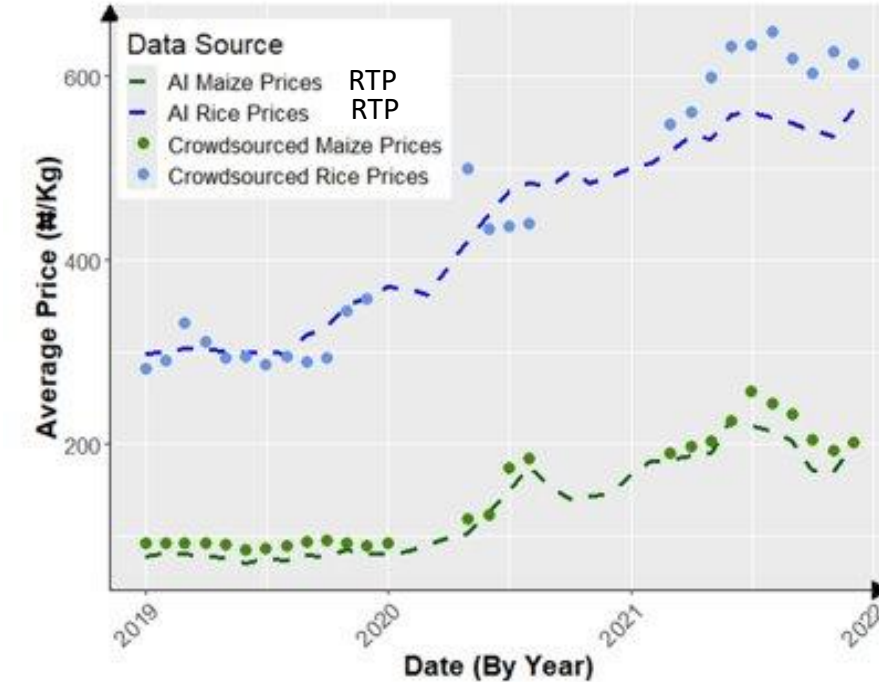
- NSA Admin



- Volunteers + Ad Hoc Staff



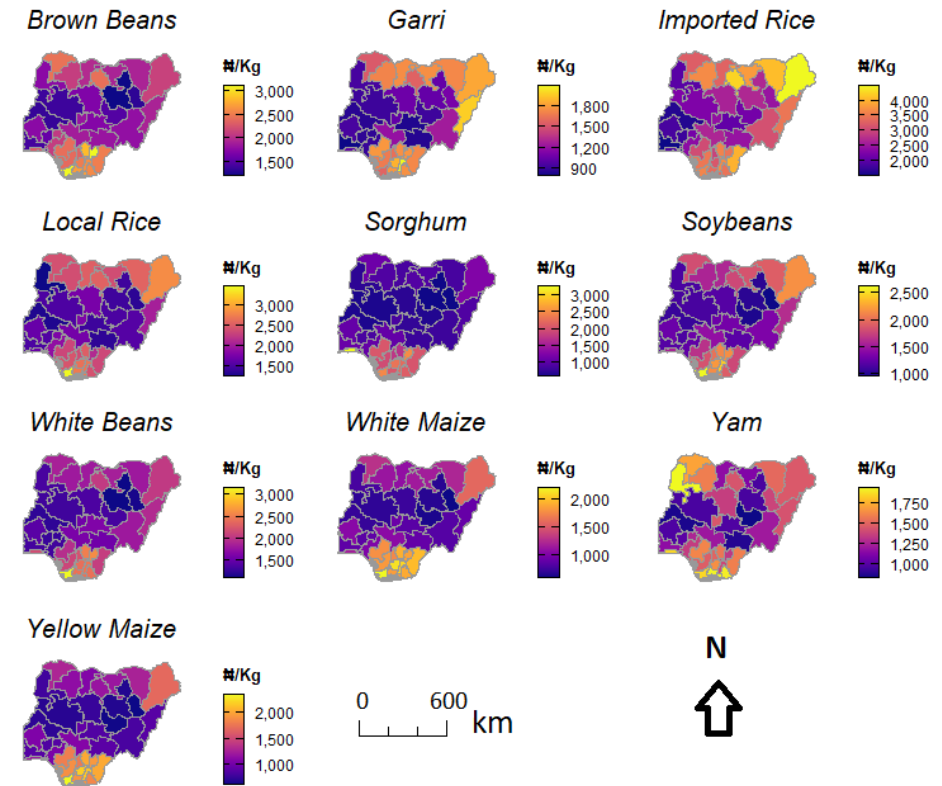
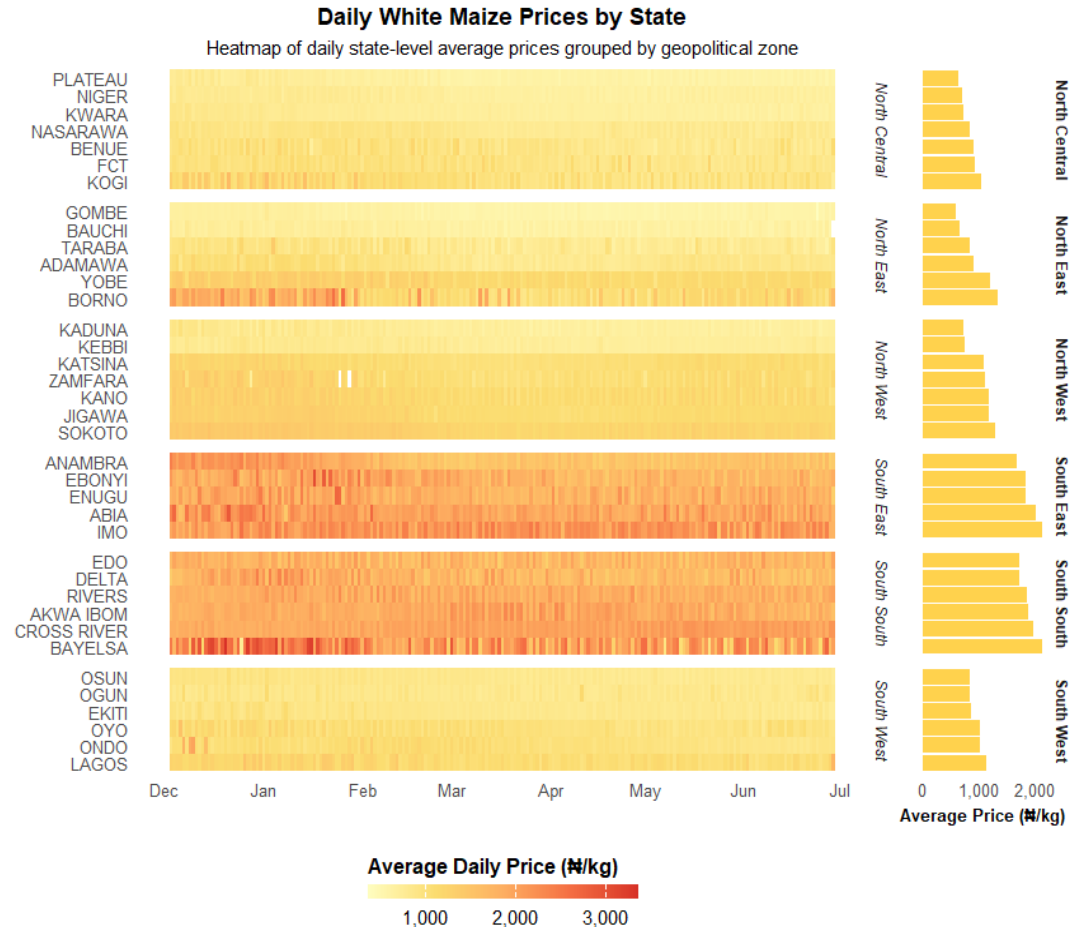
- Market Actors (Farmers/Traders)



Adewopo et al., 2025: AI-imputed and crowdsourced price data show strong agreement with traditional price surveys in data-scarce environments. PLoSOne Journal.

|| Innovating to Mitigate Data Gaps: Crowdsourcing

High Frequency Crowdsourc Price (HFCP) Data Pipeline



Average crowdsourced commodity prices from Dec 2024 – June, 2025.

Looking Forward

- Continued automation of JMR datasets, base model and dashboard for 20 countries.
- Full deployment of JMR on Global Food and Nutrition Security Dashboard.
- Progressive collaboration with technical partners, such as IPC, FAO.
- Continue data innovation to support **Food Security Preparedness Planning** to identify major crisis tipping points in near real-time



Joint Food Security Monitor - World

World, 2665 areas, 2010-01-01 - 2025-09-01, version 2025-10-09

Afghanistan, Burkina Faso, Cameroon...and 12 more, 2010 - 2025

[GET MICRODATA](#)

Reference ID WLD_2010_2025_JMR_v01_M

Producer(s) Lomme, Mathijs, Andrée, Bo Pieter Johannes

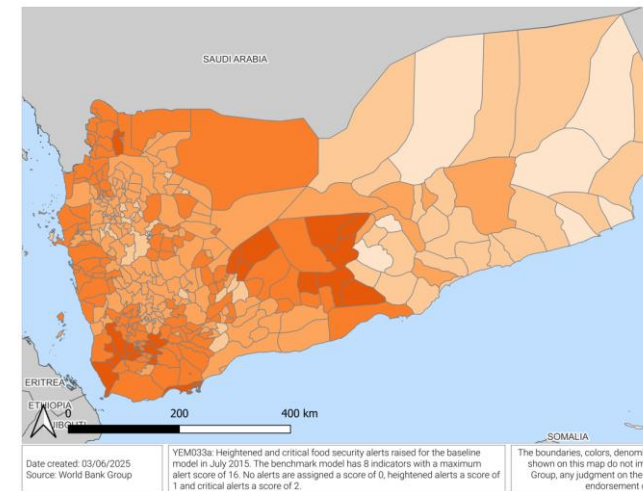
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|| Global Partners – Ongoing Technical Collaboration



|| Further Resources

- **Blog Channels**
 - Data Blog: <https://blogs.worldbank.org/en/home>
 - Agriculture and Food Blog: <https://blogs.worldbank.org/en/agfood>
- **World Bank YouTube:** <https://www.youtube.com/@WorldBank/videos>
 - JMR: https://www.youtube.com/watch?v=PnVi_3ib2Os
 - RTP: <https://www.youtube.com/watch?v=px1EeqpKDUI>
 - Preparedness Plans: <https://www.youtube.com/watch?v=gwhu64MSPyk>
- **Data 360:** <https://data.worldbank.org/>
 - Microdata Library: <https://microdata.worldbank.org/index.php/catalog/4503>
 - Data Catalog: <https://datacatalog.worldbank.org/int/search/dataset/0060165>
 - HDX (OCHA): <https://data.humdata.org/dataset/global-real-time-prices>
- **Global FNS Dashboard:** <https://www.gafs.info/>

|| Discussion and Q&A